Sergei Makeev

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Research Interests

Information retrieval, reinforcement learning, recommender systems, large language models, generative models and scaling laws, and the mathematical foundations of deep learning.

Education

2024 – 2026 **M.Sc., Applied Mathematics and Computer Science, Moscow State University**
Faculty of Computational Mathematics and Cybernetics. Diploma with Honours.
Thesis topic: Approximation of softmax activation function in recommendation problem.

2020 – 2024 **B.Sc., Applied Mathematics and Computer Science, Moscow State University**
Faculty of Computational Mathematics and Cybernetics. Diploma with Honours.
Thesis topic: Application of discrete mathematics and combinatorics in machine learning.

Work Experience

2026 – Present **Senior Researcher, AI VK**
Part of a research team led by Prof. Dr. Alexander D'yakonov, formerly world's #1 rank on Kaggle. Research topics: generative retrieval and reinforcement learning in recommender systems.

- Co-authored Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE, accepted to the 5th Workshop on End-to-End Customer Journey Optimization at KDD 2026.

2024 – 2026 **Senior Deep Learning Engineer, RecSys R&D, Yandex**
Worked under the leadership of Kirill Khrylchenko. Research topics: generative models and scaling laws, and linear attention in real-time recommendation models.

- Part of the team that developed Argus. Argus increased TLT at Yandex Music by +2.26% and like likelihood by +6.37%. It was later adapted by our colleagues to nearly every recommendation-based Yandex product (e.g., e-commerce +1.7% GMV, food-tech services +2.1% GMV, and others).
- Worked on quality-complexity trade-offs for recommendation models, co-authoring Gated Bidirectional Linear Attention for Generative Retrieval.
- Worked on pretraining large-scale generative recommender models.

2024 **Research Intern, RecSys R&D, Yandex**
Research topic: applications of graph neural networks in web-scale recommender systems.

2021 – 2024 **Research Assistant, Moscow State University**
Research topics: cognitive dynamics analysis and internal speech recognition under Russian Science Foundation grants #20-18-00067 and #20-18-00067-П, under the supervision of Dr. Alexander Vartanov.

Publications

KDD 2026 Scaling Recommender Transformers to One Billion Parameters
Kirill Khrylchenko, Artem Matveev, **Sergei Makeev**, Vladimir Baikalov

Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE, *5th Workshop on End-to-End Customer Journey Optimization*
Artem Matveev, **Sergei Makeev**, Aleksei Krasilnikov, Vladimir Baikalov, Sergei Liamaev, Kirill Khrylchenko

SIGIR 2026 Gated Bidirectional Linear Attention for Generative Retrieval
Artem Matveev, Vladislav Tytskiy, **Sergei Makeev**, Sergei Liamaev

RecSys 2025 Blending Sequential Embeddings, Graphs, and Engineered Features: 4th Place Solution in RecSys Challenge 2025
Sergei Makeev, Alexandr Andreev, Vladimir Baikalov, Vladislav Tytskiy, Aleksei Krasilnikov, Kirill Khrylchenko

Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval
Kirill Khrylchenko, Vladimir Baikalov, **Sergei Makeev**, Artem Matveev, Sergei Liamaev

Professional Service

CIKM 2026 PC member, Demo Track.

UMAP 2026 PC member, Industry Track.

UMAP 2025 PC member, Industry & Start-up Track.

Patents

Dec 2025 Application for invention 2025136959, Federal Service for Intellectual Property
Yandex filed a patent application for Argus (see our paper Scaling Recommender Transformers to One Billion Parameters).

Competitions & Olympiads

Jun 2025 RecSys Challenge 2025 (RecSys'25)
Led team `ambitious` to 4th place.

Apr 2024 Moscow State University, The Lomonosov Universiade in Mathematics and Computer Science
2nd-degree diploma.

Apr 2023 Moscow State University, The Lomonosov Universiade in Mathematics and Computer Science
1st-degree diploma.

Talks

Sep 2025 Talk at RecSys'25, O2 Universum, Prague, Czech Republic
Presented my team's solution to the RecSys Challenge 2025 as an invited speaker.

Mar 2026 Talk at HSE University, Moscow, Russia
Gave a guest lecture for the DeepRecSys course taught by Kirill Khrylchenko at the Faculty of Computer Science, HSE University.

Teaching

- Contributed to the DeepRecSys undergraduate course taught by Kirill Khrylchenko at the Faculty of Computer Science, HSE University.

Research in News and Social Media

- Argus received broad attention after its launch. It was featured in news outlets (e.g., ict.moscow, yagla.ru, mltimes.ai, setka.ru, toolarium.ru, m.seonews.ru, rozetked.me, pikabu.ru), covered on habr.com, and discussed on both x.com and habr.com. It was later adapted by our colleagues to many other Yandex products. Its deployment in Yandex Ads also drew extensive discussion (e.g., timeweb.com, habr.com, adindex.ru, habr.com, dyvannyi-it.ru, marketplacegu.ru, sostav.ru, etc.). Sumit Kumar, author of a popular information retrieval blog (4.1k subscribers on x.com as of Jul 2026), wrote about Argus on x.com and listed it in Top Information Retrieval Papers of the Week.
- Our paper Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval also received extensive discussion: vc.ru, ict.moscow, kommersant.ru, naked-science.ru, cnews.ru, itsnew.ru, etc. It was listed among Top Information Retrieval Papers of the Week.
- Gated Bidirectional Linear Attention for Generative Retrieval, which I co-authored, was also noted by the community on linkedin.com and x.com. It was listed in Sumit Kumar's popular blog Top Information Retrieval Papers of the Week.
- Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE was also noted by Sumit Kumar on x.com and included in his Top Information Retrieval Papers of the Week rating. After its release, it was promptly cited by some scientific websites (e.g., gist.science, moltbook.com, chatpaper.com).

Mentorship

- Mentored two interns at Yandex, Oleg Sorokin and Vladislav Dodonov. Both are now core developers on the RecSys R&D team (see, for example, Gryphon).

Grants

- Worked on cognitive dynamics analysis and internal speech recognition under Russian Science Foundation grants #20-18-00067 and #20-18-00067-П as a research assistant at Moscow State University, under the supervision of Dr. Alexander Vartanov.

Technical Skills

Languages	Current: Python. Past: C/C++, Assembly, CUDA.
Frameworks	Current: PyTorch. Past: TensorFlow, JAX.